SC7 Bayes Methods MT26

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Lecture 1: outline of Bayesian inference

Lecture Notes and Problem sheets are available at

`https://canvas.ox.ac.uk/courses/298238`

Course outline

The following is aspirational. The topics and sequence are correct. However, timing may vary.

Week 1 - (1) Bayesian inference, (2) Bayes example

Week 2 - (3) Marginal consistency and de Finetti's Theorem,
(4) Utility and coherent inference

Week 3 - (5) The Savage Axioms, (6) MCMC

Week 4 - (7) Data Augmentation,
(8) Estimating Bayes factors, Laplace Approximation

Week 5 - (9) ABC*, (10) Model Averaging

Week 6 - (11-12) Jacobian and Reversible Jump MCMC

Week 7 - (13) RJ example (14) Bayesian non-parametrics

Week 8 - (15-16) Dirichlet Process and DP example

*Approximate Bayesian Computation

Bayesian inference

Some unknown real-world quantity Θ takes values in Ω. Typically Ω is some p-dimensional subset of R^p below.

Let $S \subseteq \Omega$. Is Θ in S? We have some knowledge that may help us answer this question, and we will additionally gather data.

If our knowledge is coherent then it can be represented by a prior distribution ($\pi(\theta),\ \theta \in \Omega$ say).*

$$\int_S \pi(\theta) d\theta = \Pr(\Theta \in S).$$

The distribution $\pi(S) = \Pr(\Theta \in S),\ S \in \mathcal{B}_\Omega$ represents our state of knowledge before we see the data.

*For prior existence and uniqueness see Savage and de Finetti weeks 2&3.

Observation model and generative model

Observations $Y = (Y_1, ..., Y_n), Y \in \mathcal{Y}$ have density $p(y|\theta)$ for $Y = y$ given $\Theta = \theta$.

We have given the following Generative Model

Algorithm 1 *Simulation of* Θ, Y.

1. *simulate* $\theta \sim \pi(\cdot)$
2. *simulate* $y \sim p(\cdot|\theta)$
3. *return* $(\Theta = \theta, Y = y)$.

The joint distribution of y, θ is $p(y|\theta)\pi(\theta)$. Conditioning on y,

$$\pi(\theta|y) \propto p(y|\theta)\pi(\theta)$$

the Posterior density. Is this what nature did? Where does the prior come from? I have a posterior distribution, now what?

The Bayesian Belief Update

We observe $Y = y$. How do our beliefs about $\Theta \in S$ change?

$$\Pr(\Theta \in S | Y = y) = \int_S \pi(\theta|y) d\theta$$

is the posterior probability the unknown true Θ lies in S, given the data and the prior knowledge expressed in $\pi(\theta)$.

The posterior distribution $\pi(S|y) = \Pr(\Theta \in S | Y = y),\ S \in \mathcal{B}_\Omega$ represents our state of knowledge after we see the data.

The normalised posterior is

$$\pi(\theta|y) = \frac{p(y|\theta)\pi(\theta)}{p(y)}$$

with $p(y) = \int_\Omega p(y|\theta)\pi(\theta)d\theta$ the normalising marginal likelihood.

Model selection and goodness-of-fit

$p(y)$ is also the prior predictive density for the data.

We prefer generative models that predict the data well, so we prefer models with the larger $p(y) \rightarrow$ Bayes factors[†].

The posterior predictive distribution for new data, with density

$$p(y'|y) = \int_\Omega p(y'|\theta)\pi(\theta|y)d\theta$$

is useful for prediction and model selection with held-out data.

Also in goodness of fit checks. Our model should predict new data that resembles old: simulated $y' \sim p(\cdot|y)$ resembles y.

For example if $g : \mathcal{Y} \rightarrow R$ then $\Pr(g(Y') < g(Y)|Y = y)$ shouldn't be too small - like a p-value in a GOF test.

[†]see Kass and Raftery (1995) for Bayes factors and Berger and Pericchi (1996) for model selection with $p(y'|y)$ using intrinsic Bayes factors.

Summarising the posterior

When we plan the inference we have some goal, eg estimate the true value of a parameter.

In general our "actions" in carrying out "inference" should be guided by Decision Theory. Savage calls this coherent inference.

We elicit a loss expressing our objective, eg. $l(\hat{\theta}, \theta^*) = (\hat{\theta} - \theta^*)^2$ if we estimate $\hat{\theta}$ when the truth is θ^*.

We estimate the value minimising the Expected Posterior Loss

$$\hat{\theta} = \arg\min_\theta E\{l(\theta, \Theta)|Y = y\}.$$

In the example

$$E\{(\theta - \Theta)^2|y\} = \mathrm{var}(\Theta|y) + (\theta - E\{\Theta|y\})^2$$

so $\hat{\theta}$ is the posterior mean $E(\Theta|y)$.

If $\theta^{(t)} \sim \pi(\cdot|y),\ t = 1, ..., T,\ f(\theta) = \theta$ and

$$\bar{f}_T = \frac{1}{T}\sum_{t=1}^{T} f(\theta^{(t)}) \qquad \text{then } \hat{\theta} = \bar{f}_T \text{ estimates } \mu(y).$$

Similarly if $f(\theta) = \mathbb{I}_{\theta \in S}$ then $\bar{f}_T$ estimates $\pi(S|y)$.

The level-α HPD credible set C_α for θ satisfies

$$\int_{C_\alpha} \pi(\theta|y)d\theta = 1 - \alpha,$$

so $\Pr(\Theta \in C_\alpha|Y = y) = 1 - \alpha$, with the additional constraint

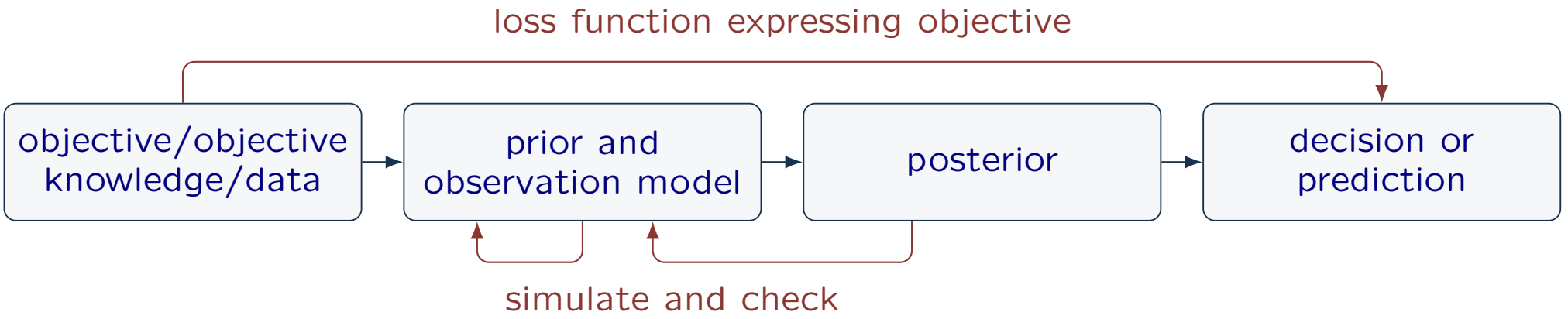
$$\theta \in C_\alpha \text{ and } \theta' \in \Omega \setminus C_\alpha \ \Rightarrow\ \pi(\theta|y) \geq \pi(\theta'|y),$$

so higher posterior density within the set. These can be estimated from Monte Carlo samples $\theta^{(t)} \sim \pi(\cdot|y),\ t = 1, ..., T$.

The HPD also has a justification from Decision Theory[‡].

[‡]See Exercise 1.6-1,7 in notes Section 1 Appendix C for these claims.

The Bayesian inferential pipeline



We start with knowledge of the world and data. We have a question and objective (the loss or price we pay for answering wrong).

Designing a prior and observational model is statistical modeling.

Summarising the posterior is computational statistics.

Coherent Bayesian inference reaches decisions by minimising the expected posterior loss. That is decision theory.

Prior Elicitation Think about the prior! When Bayes leads to nonsense, it is usually caused by careless prior specification.

The issue is obviously important when the data are only weakly informative of the parameter.

The problem of prior specification is acute in high dimensional problems. Metric factors in uniform priors are a common trap.

Example: suppose our parameter $\theta \in \Omega$ takes values in a p-dim ball of radius r, $\Omega = \{\theta \in R^p : |\theta| \leq r\}$ with volume

$$V_p(r) = \pi^{p/2} r^p / \Gamma(1 + p/2)$$

and we take $\pi(\theta) = V_p(r)^{-1}$, a uniform prior.

The prior probability $\Pr(r - \delta \leq |\Theta| \leq r) \to 1$ as $p \to \infty$ as

$$\frac{\text{Shell volume}}{\text{Total volume}} = 1 - (1 - \delta/r)^p.$$

So for large p our prior is strongly informative of $|\theta|$.

Prior elicitation checklist

- Is the parameter θ generated by some process we can model? The distribution over θ determined by the process *is* the prior.

- Do the parameters correspond to "elements of reality"? That makes it easier to identify prior knowledge.

- Is there some physically interpretable function of the parameter $f(\theta)$? Check prior samples realise a priori plausible f-values.

- The prior should in general be non-informative with respect to any key scientific hypothesis or parameter.

- The prior you write down should express prior knowledge. Simulate and check prior samples are distributed as intended.

- We typically check results are insensitive to a range of priors representing different states of knowledge.

Model selection and model averaging

Let $\mathcal{M}$ be a set of model index labels $m = 1, 2, ..., |\mathcal{M}|$.

For model m we have a prior $\pi(\theta|m), \theta \in \Omega_m$ and an observation model $p(y|\theta, m), y \in \mathcal{Y}$. The parameter space may vary across models. Taken together they give the generative model

$$(\theta, y) \sim \pi(\theta|m)p(y|\theta, m)$$

("the model"). We choose between different generative models.

The posterior density under this model is

$$\pi(\theta|y, m) = \frac{p(y|\theta, m)\pi(\theta|m)}{p(y|m)}, \qquad \text{with}$$

$$p(y|m) = \int_{\Omega_m} p(y|\theta, m)\pi(\theta|m)d\theta$$

the marginal likelihood under model m.

If M is the unknown true model, and $M \in \mathcal{M}$, then treat M as an unknown parameter like Θ and use the same pipeline, with

$$\pi_M(m) = \Pr(M = m)$$

our prior probability for m to be the true model. The model posterior $\Pr(M = m|Y = y) = \pi_M(m|y)$ is

$$\pi_M(m|y) = \frac{p(y|m)\pi_M(m)}{p(y)}$$

and

$$p(y) = \sum_{m \in \mathcal{M}} p(y|m)\pi_M(m)$$

is the overall marginal likelihood, now averaged over models. The model averaged posterior* allows for uncertainty in the model

$$\pi(\theta|y) = \sum_{m \in \mathcal{M}} \pi(\theta|y, m)\pi_M(m|y).$$

*Section 8 of notes. We assume $\pi(\theta|y, m)$ is defined on each Ω_m.

Why do model selection?

Model Construction/Improvement - We have a few models and we want to find the one that best fits the data.

Goodness of fit/Model expansion - We define a model M1 incorporating likely model extensions and compare M0 and M1.

Hypothesis testing for θ - two scientists have different beliefs, for eg $\theta \in S_1,\ \theta \in S_2$. Which do the data and model support?

Hypothesis testing for M - which generative model $\pi(\theta|M = m)p(y|\theta, M = m)$ is the true model for the data?

Why we shouldnt do model selection[†]

If we select a model, fix it and then do parameter estimation, we dont allow for uncertainty in the model $\Rightarrow$ model averaging.

[†]As we show in Section 8.1.2 of the notes.

How to do model selection in principle

Suppose we choose the model with the largest posterior probability[‡]. If m and m' are two models then we favor m if

$$A_{m,m'} = \pi_M(m|y)/\pi_M(m'|y) > 1.$$

The posterior odds $A_{m,m'}$ have a simple meaning. Model m is $A_{m,m'}$ times more probable a posteriori than model m'.

We may be concerned that our prior weighting $\pi_M(m)$ is distorting this ratio. The Bayes factor

$$B_{m,m'} = p(y|m)/p(y|m')$$

equals $A_{m,m'}$ if $\pi_M(m) = \pi_M(m')$, the prior weighting is equal.

[‡] $l(\hat{m}, m^*) = \mathbb{I}_{\hat{m} \neq m^*}$ so $E(l|Y = y) = 1 - \pi_M(\hat{m}|y)$ and mode minimises EPL.

Marginal likelihoods, Posterior odds and Bayes factors have a built in penalty on complexity.

In about lecture 8 we will show

$$\log(p(y|m)) = \ell(\hat{\theta}; y) - (p/2)\log(n) + O(1)$$

where $\hat{\theta}$ is the MLE and p is parameter dimension.

As the prior becomes more diffuse, or high dimensional, the probability mass it puts on the support of the likelihood goes down, so, unless there is a compensating increase in likelihood, $p(y|m)$ tends to decrease with increasing model complexity.

$p(y|m)$ is the average of $p(y|\theta, m)$ over samples from $\pi(\theta|m)$ so it depends on $p(y|\theta, m)$ and $\pi(\theta|m)$ everywhere in Ω_m

$\Rightarrow$ model mispecification anywhere (even in the tails, at physically uninteresting values of θ) distorts $B_{m,m'}$.

$\Rightarrow$ If we are using priors to represent hypotheses we have to be careful the tail behavior is supported by prior knowledge.[§]

Marginal likelihoods are hard to estimate and good priors hard to construct. These are the main obstacles to Bayesian inference.

[§]This leads to the Lindley paradox.